Bt405 current solved papers.

Q:1 Current use of nanbiotechnology?.)5 marks

Ans: Nanotechnology: Current Uses

Nanotechnology is not an industry; nor is it a single technology or a single field of research

The field of nanotechnology is interdisciplinary.

Nanotechnology is referred to as a general-purpose technology.

The total market for nanobiotechnology products was $19.3 billion in 2010 and is growing at a

Compound annual growth rate (CAGR) of 9% to reach a forecast market size of $29.7 billion by

2015.

Medical applications, including drug delivery and microbicides, dominate today's market, with

Sales of $19.1 billion in 2010. This market segment is growing at a compound annual growth

Rate (CAGR) of 8.7%, and is forecast to reach sales of $29 billion by 2015 (BCC).

More and more consumer products are branded worldwide as nano. Nano-sized TiO2 and ZnO – sunscreens

Silver nanoparticles – antibacterial properties (soaps, toothpastes, deodorants, lip products,

Make-up instruments, hair brushes, curling tongs, foils for electric razors, foot massagers, tooth

Brushes, rubber gloves, hair dryers, facial ionic steamers).

Nanosomes – improve the solubility of ingredients and add shimmer.

Nano emulsions and nanosomes -preserve active ingredients, such as vitamins and anti-oxidants.

Nano gold – healing and anti-oxident properties.

Fullerenes, bucky balls – anti-oxidant and smoothing properties in moisturisers, increased

Penetration into the skin , anti-aging properties.

Q:2 Self healing adhesive and why they serves as biomimetic model for self healing materials?.) 5 marks

Ans: Diatoms are a type of algae with nanostructured amorphous silica surfaces.

 Diatoms are unicellular algae that have the capability to colonize any natural and manmade submerged surfaces.

 Diatoms are a large group of eukaryotic microalgae that have a rigid cell wall made of

Silica (SiO_2).

 Diatoms are often among the first organisms to colonize underwater surfaces, because

Their adhesive mucilage enables the cells to attach to substrata of a very wide range of

Chemical composition (organic, inorganic, natural or man-made).

 Surface adhesion of diatoms belonging to the raphid pennate type requires the secretion

Of so-called adhesive mucilage strands through a dedicated slit in the silica cell wall,

Which is termed the raphe.

 Adhering cells can then move across the surface, which is accomplished through an actinmyosin dependent cytoskeletal motor that translocates the adhesive mucilage strands in a

Rearward direction through the raphe, thereby moving the cell forward

Some diatom species have evolved strong self-healing underwater adhesives.

Some are free-floating, others have adhesive properties in water: for instance, diatoms in the

Antarctic seas can attach to ice.

 Others secrete viscous mucilage which binds colonies together while protecting the silica

Shells from wear as they rub against each other.

 Molluscs are another species that have adhesive properties under water.

 Both diatoms and molluscs have strong underwater glues that can also resist stress and

Self-heal.

 They serve as a biomimetic model for self-healing materials.

 These natural adhesives and their self-healing properties are due to the properties of the

Proteins contained in them.

 These proteins have sacrificial bonds that allow the molecule to be reversibly stretched

By breaking and re-bonding.

 This sacrificial bond behaviour has been observed in many other materials, such as wool.

 Closing the wounds with sticky materials has been introduced to prevent bleeding and

Induce the wound healing process.

 Many types of surgical adhesives including tissue adhesives have been developed to be a

Suitable alternative for sutures and staples.

 A new approach which aims at producing bio-adhesives by mimicking the nature along

With applying nanotechnology methods.

 Engineering stable underwater adhesives currently poses a major technical challenge,

Because most man-made adhesives fail in wet conditions, owing to chemical modification

Of the adhesive or its substrate.

 Diatoms produce adhesives that are extremely strong and robust, in both fresh- and

Seawater environments.

 AFM (Atomic Force Microscopy) study of living diatoms, phase imaging and force spectroscopy experiments revealed characteristics of these natural adhesives that may be of use in designing man-made adhesive analogues that function in wet environments.

 The AFM allows for investigations of micromechanical properties of the cell surface, for example viscoeleastic properties, adhesion forces and hardness measurements, in physiological conditions

Experiments with three diatom species that produce outstanding natural

Adhesives: Eunotia sudetica, Navicula seminulum and an unidentified species

Q:3 5 ways nanotechnology impacts our lives daily basis?.) 3 marks

Ans: 1. Faster, smaller, and more powerful computers that consume far less power, with longerlasting batteries.

2. Faster, more functional, and more accurate medical diagnostic equipment. Lab-on-a-chip

Technology enables point-of-care testing in real time, which speeds up delivery of

Medical care. Nanomaterial surfaces on implants improve wear and resist infection.

3. Nanoparticles in pharmaceutical products improve their absorption within the body and

Make them easier to deliver, often through combination medical devices. Nanoparticles

Can also be used to deliver chemotherapy drugs to specific cells, such as cancer cells.

4. Improved vehicle fuel efficiency and corrosion resistance by building vehicle parts from

Nanocomposite materials that are lighter, stronger, and more chemically resistant than

Metal. Nanofilters remove nearly all airborne particles from the air before it reaches the

Combustion chamber, further improving gas mileage.

5. Nanoparticles or nanofibers in fabrics can enhance stain resistance, water resistance, and

Flame resistance, without a significant increase in weight, thickness, or stiffness of the

Fabric. For example, “nano-whiskers” on pants make them resistant to water and stains.

Q :4 self assembled mono layers?. 3 marks

Ans: Developed by the oil industry, MCM-41 (known also as “self-assembled monolayers on

Mesoporous supports,” SAMMS), with pore sizes in the range of 10 – 100 nanometers, is used

For the removal of ultrafine contaminants.

Q:5 application of SAM? 2 marks

Ans: Applications: SAMs are extremely useful in thin-film technology since they provide a

Simple, versatile and relatively inexpensive way of producing coatings with specific

Functionalities.

 Thin coatings are used in many applications, for example to control surface wetting and

Adhesion properties; to provide chemical resistance or specific biocompatibility; to confer

Antibacterial properties; and for sensors.

 SAMs are particularly interesting for developing DNA and protein micro and nanoarrays

For medical screening and for future electronic components in computer chips and other

ICT developments.

Q:6 intermolecular and intramolecular bonding differwnce ?2 marks

Ans: Intramolecular bonding (chemical interactions): these are bondings that involve

Changes in the chemical structure of the molecules and include ionic, covalent and

Metallic bonds;

 Intermolecular bonding (physical interaction): these are bondings that do not involve

Changes in the chemical structure of the molecules and include ion-ion and ion-dipole

Interactions; van der Waals interactions; hydrogen bonds; hydrophobic interactions;

Repulsive forces (such as steric repulsions).

BT 405 12:00 pm

1-applications of self assembled monomers

2-nanotechnology

Ans: Simplest _ technology at the nonscale

 Nanotechnology literally means any technology on a nanoscale that has applications in

The real world.

 Succinct definition of nanotechnology is engineering with atomic precision or atomically

Precise technology _ APT

3-Environmental application of metal nano partical

Ans: Some forms of metal nanoparticles have important environment applications.

 Zero-valent (Fe0) iron nanoparticles are under investigation for the remediation of

Contaminated groundwater and soil.

 Iron, when exposed to air, oxidises easily to rust; however, when it oxidises around

Contaminants such as trichloroethylene (TCE), carbon tetrachloride, dioxins, or PCBs,

These organic molecules are broken down into simple, far less toxic carbon compounds. Iron nanoparticles are more effective than conventional ‘iron powder’, which is already

Used to clean up industrial wastes. Iron nanoparticles are 10 to 1 000 times more reactive

Than commonly used iron powders.

 Silver nanoparticles have a strong antibacterial capacity.

 They are used in numerous products to prevent or reduce the adherence of bacteria to

Surfaces.

4-role of nanotechnology in diagnosis

Ans: Several nanoparticles have been used for diagnostics. Of these, the most frequently used

Are gold nanoparticles, QDs, and magnetic nanoparticles.

 Nanodiagnostics is the new term that describes use of methods and techniques of

Nanotechnology and its principles for diagnostics purposes.

 It includes, although not limited to, the manipulation and assessment of single molecule,

Size reduction of systems and platforms to make use of nanoscale properties obtainable

From interactions between surfaces and biomolecules.

 Nanodiagnostics is an evolving application of nanoscale technology to meet the demand

Of clinical diagnostics, determining disease state, any predisposition to such, the

Pathology of the condition and the identification of causative organisms

 With nanotechnology, diagnosis is being carried out on a nano-scale leading to a trend of

The use of hand held devices that are easy to use and marketable.

5-self healing adhesive & role of diatom

6-role of nanotechnology in medicine.

Ans: Diagnostic

- Imaging
- Quantum dots
- Microscopic sampling
- Detection of airway abnormalities

Therapeutic

- Delivering medication to the exact location
- Killing of bacteria, viruses & cancer cells
- Repair of damaged tissues
- Oxygen transport

- Skin and dental care

- Augmentation of immune system

- Treatment of Atherosclerosis

- The clottocyte concept

- Brain enhancement